Window Functions Assignment Questions

1. What is the main difference between ROW_NUMBER() and RANK() functions in SQL?

A) ROW_NUMBER() generates unique values, while RANK() generates gaps in ranks.
B) ROW_NUMBER() allows duplicates, while RANK() does not.
C) ROW_NUMBER() resets after every partition, while RANK() continues.
D) Both functions generate the same result set.

Correct Answer: **A**

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2. Which function ensures that ranks are consecutive, even when there are ties?

A) ROW_NUMBER()
B) RANK()
C) DENSE_RANK()
D) NTILE()

Correct Answer: **C**

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3. If a dataset contains three rows with the same value, how will the RANK() function assign ranks to those rows?

A) All three rows will get the same rank, and the next row will skip two ranks.
B) The rows will get consecutive ranks.
C) Only the first row gets ranked, and the others are skipped.
D) None of the rows will be ranked.

Correct Answer: **A**

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4. Which function generates a unique rank for every row, even when values are the same?

A) ROW_NUMBER()
B) RANK()
C) DENSE_RANK()
D) PERCENT_RANK()

Correct Answer: **A**

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5. In which scenario will DENSE_RANK() and RANK() produce the same result?

A) When all the values in the result set are unique.
B) When the result set contains duplicate values.

C) When ROW_NUMBER() is used with DENSE_RANK().
D) When partitioning is applied in the result set.

Correct Answer: **A**

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6. If the dataset contains two rows with the same value and DENSE_RANK() is applied, what will be the rank of the next row?

A) It will skip one rank.
B) It will continue with the next consecutive rank.
C) It will skip two ranks.
D) It will repeat the previous rank.

Correct Answer: **B**

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7. Which function is most appropriate to rank items without any gaps in the ranking sequence, even in the case of ties?

A) ROW_NUMBER()
B) RANK()
C) DENSE_RANK()
D) NTILE()

Correct Answer: **C**

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8. What is the output of ROW_NUMBER() in the presence of duplicate values?

A) It assigns the same rank to duplicate values.
B) It assigns unique sequential numbers to each row, regardless of duplicates.
C) It skips ranks for duplicate values.
D) It ignores duplicate values entirely.

Correct Answer: **B**

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9. Which function will skip ranks when there are ties in the ranking values?

A) ROW_NUMBER()
B) RANK()
C) DENSE_RANK()
D) NTILE()

Correct Answer: **B**

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10. When using RANK() in SQL, what happens if two rows have the same rank?

A) The function skips to the next rank number after assigning ranks to the tied rows.
B) The function assigns consecutive ranks to the tied rows.

C) The function removes the duplicate rows from the result set.
D) The function repeats the rank and continues with the next rank without skipping.

Correct Answer: **A**

SQL Query Questions

1. Rank Vendors by Total Sales

Question: Rank vendors by their total sales amount, with the highest sales ranked first, and display the rank alongside each vendor.

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2. Assign Row Numbers to Each Customer's Purchase

Question: Assign a row number to each customer's purchases, ordered by the market_date, and display the row numbers alongside the customer details.

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3. Rank Products by Price for Each Vendor

Question: Rank all products sold by each vendor by price in descending order and display the rank for each product.

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4. Find Vendors with the Highest and Lowest Product Prices

Question: Use ROW_NUMBER() to find the vendor with the highest and lowest-priced products for each product category.

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5. Assign DENSE_RANK to Vendors Based on Booth Price

Question: Assign a dense rank to vendors based on the price level of the booths they are assigned and display the rank along with vendor details.

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6. Rank Customers by Total Purchase Amount

Question: Rank customers by their total purchase amount, with the highest purchase amounts ranked first, and display the rank alongside each customer.

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7. Assign a Row Number to Each Product Sold in Each Booth

Question: Assign a row number to each product sold in each booth, ordered by market_date, and display the row number for each product.

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8. Rank Products by Total Quantity Sold

Question: Rank products by the total quantity sold across all vendors using RANK(), and display the rank alongside the product details.

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9. Use ROW_NUMBER() to Find the Third Highest Revenue Vendor

Question: Use ROW_NUMBER() to find the vendor with the third-highest total revenue from product sales.

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10. Assign DENSE_RANK to Customers Based on Purchase Frequency

Question: Assign a dense rank to customers based on the frequency of their purchases, with the highest frequency ranked first, and display the rank alongside customer details.